

Statistics & Probability

Week 1 Assignment

Instructions

1. Read the problem statements carefully.
2. Identify the concept in which this problem fits into and state the concept.
3. Solve the problem.
4. Detail out the steps you have taken to solve this. Do not give only answers.

1. Consider the Bernoulli distribution with $p=0.8$. If a sequence of 10 trials are done, what is the expected value of the number of successes that occur?

2. A company makes electronic components for TV's.
95% pass final inspection (and 5% fail and need to be fixed).
120 components are inspected in one day. What is the expected
number that fail in one day?

3. The ages of the population of a town are Normally distributed with mean 43 and standard deviation 14. The town has a population of 5,000.

How many would you expect to be aged between 22 and 57?

4. A lens outlet owner buys his glass from four different manufacturers - Clearglass (10%), Strongpane (25%), Mirrorglass (30%) and Reflection (35%).

In the past, the outlet owner has found that 1% of Clearglass' product is cracked, 1.5% of Strongpane's product is cracked, and 2% of Mirrorglass' and Reflection's products are cracked.

He removes the protective covering from a sheet of glass without looking at the manufacturer's name - in other words, it's a random choice. He finds the glass is cracked. What is the probability it was made by Mirrorglass?

5. In a group of 25 boys, 20 play ice hockey and 17 play baseball. They all play at least one of the games. What is the probability that a boy chosen at random from the class plays ice hockey but not baseball?

6. In a class of 35 children, 22 like bananas, 18 like cherries and 13 like strawberries. 7 of them like bananas and cherries. 8 of them like bananas and strawberries. 5 of them like cherries and strawberries.

They all like at least one of the fruits.

What is the probability that a child chosen at random from the class likes cherries only?

7. Two fair dice are thrown.

What is the probability that the score on the first die is 6 or the score on the second die is 5?

All the best!!!